LIFTING A MAP FROM A STRICTLY PERFECT COMPLEX THROUGH A QUASI-ISOMORPHISM, LOCALLY UP TO HOMOTOPY

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file sites-cohomology.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The proof is the author's complete argument, ending with the author's own closing phrase, which is retained verbatim.

Editorial note. The Xy-pic diagram of the counted statement is delivered as the author wrote it, including the dotted arrow.

Editorial note. Cross-chapter restatement: the same label occurs in chapter cohomology for a ringed space. That statement is the special case of this ringed-site statement, so per the lane-s4 rule only this more general occurrence is counted here.

Editorial note. The single labelled reference occurring inside the counted spans is carried in assembly.external_references as a non-load-bearing navigation pointer to the same chapter.

Editorial note. The preamble is the chapter's own retained minimal preamble (sources/stacks-sites-cohomology/retained-preamble.tex), a verbatim excerpt of the native preamble: the Xy-pic setup and the native theorem declarations with the mathematical macros.

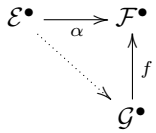
Allowed standard prerequisites (editorial).

- Strictly perfect complex of O_U-modules. Same chapter, the definition of a strictly perfect complex as a bounded complex of direct summands of finite free modules. Supplies the hypothesis on E^bullet that makes the local triviality lemma applicable.
- Cone of a morphism of complexes and its cohomology. Same chapter, the cone C(f)^bullet of a morphism f and the associated distinguished triangle. Supplies the reduction of the hypotheses on H^j(f) to the vanishing of the cohomology of the cone in degrees at least a.
- Distinguished triangles in the homotopy category K(O_U). Same chapter, the triangle G^bullet -> F^bullet -> C(f)^bullet -> G^bullet[1] and the long exact sequence of homotopy classes it induces. Supplies the final step, where vanishing of the composite is converted into a lift up to homotopy.

Source reference. Original lemma 08FP: sources/stacks-sites-cohomology/pinned-original-source.tex, source lines 11796–11796. Same-chapter navigation pointer invoked at proof line 11866, where the author applies the local triviality of maps from a strictly perfect complex to a complex with vanishing cohomology above a. The pointer only resolves that reference for the reader; the cited lemma is not reproduced inside the counted span, and no bibliography citation is involved.

Date: stacks-project:sites-cohomology.tex:local-snapshot-20261011; GNU FDL 1.2 or later.

Lemma 0.1. *Let $(\mathcal{C}, \mathcal{O})$ be a ringed site. Let U be an object of $\mathcal{C}$. Given a solid diagram of complexes of $\mathcal{O}_U$-modules*



with $\mathcal{E}^\bullet$ strictly perfect, $\mathcal{E}^j = 0$ for $j < a$ and $H^j(f)$ an isomorphism for $j > a$ and surjective for $j = a$, then there exists a covering $\{U_i \to U\}$ and for each i a dotted arrow over U_i making the diagram commute up to homotopy.

Proof. Our assumptions on f imply the cone $C(f)^\bullet$ has vanishing cohomology sheaves in degrees $\geq a$. Hence Lemma 08FP guarantees there is a covering $\{U_i \to U\}$ such that the composition $\mathcal{E}^\bullet \to \mathcal{F}^\bullet \to C(f)^\bullet$ is homotopic to zero over U_i. Since

$$\mathcal{G}^\bullet \to \mathcal{F}^\bullet \to C(f)^\bullet \to \mathcal{G}^\bullet[1]$$

restricts to a distinguished triangle in $K(\mathcal{O}_{U_i})$ we see that we can lift $\alpha|_{U_i}$ up to homotopy to a map $\alpha_i : \mathcal{E}^\bullet|_{U_i} \to \mathcal{G}^\bullet|_{U_i}$ as desired. □